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United States Patent Application Publication

20250282552

Kind Code

A1

Publication Date

September 11, 2025

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CONTAINER UNLOADING APPARATUS

Abstract

An apparatus for emptying the contents of a container. The apparatus may include a base, a frame, and a vertical lift. The lift may include a drive portion fixed to the base and a linearly movable member pivotally coupled to the frame. The drive portion may be configured for driving the linearly movable member for movement relative to the base along a substantially linear axis. The lift may also include a pivot arm having a first end coupled to the base and a second end pivotally coupled to the frame. As the drive portion drives the linearly movable member along the substantially linear axis the frame moves upwardly and rotates relative to the base. The apparatus may include a base and a frame including an extension frame slidably movable between a non-extended position and an extended position. The apparatus may also include a conveyor disposed adjacent the extension frame.

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Family ID: 1000008529062

Appl. No.: 19/072466

Filed: March 06, 2025

Related U.S. Application Data

us-provisional-application US 63562349 20240307

us-provisional-application US 63648873 20240517

Publication Classification

Int. Cl.: B65G7/08 (20060101); B65D88/56 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of the filing date of U.S. Provisional Application Ser. No. 63/562,349, filed Mar. 7, 2024, and U.S. Provisional Application Ser. No. 63/648,873, filed May 17, 2024, the teachings of which applications are hereby incorporated herein in their entirety.

TECHNICAL FIELD

[0002] The technical field relates to management of containers and, more specifically, to a container unloading apparatus with a vertical lift.

BACKGROUND INFORMATION

[0003] Containers are often used in industrial applications to manage large amounts of inventory. These containers are often filled with like pieces such as, for example, manufactured parts, raw materials, mail, goods, etc. Because these containers are heavy and unwieldy, loaders and unloaders are often specialized to facilitate loading and unloading of the containers. One concern with a container loader/unloader is the physical space it occupies when in an unloading position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Objects, features, and advantages disclosed herein will be apparent from the following description of particular embodiments, as illustrated in the accompanying drawings in which like reference characters/descriptions refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles disclosed herein. For purposes of clarity, not every component may be labeled in every drawing.

[0005] FIG. 1A is a right rear isometric view of one example embodiment of a container unloading apparatus consistent with the present disclosure wherein the apparatus is in a loading position.

[0006] FIG. 1B is a left rear isometric side view of the container unloading apparatus shown in FIG. 1A.

[0007] FIG. 1C is a left side view of the container unloading apparatus shown in FIG. 1A.

[0008] FIG. 1D is a top view of the container loading apparatus shown in FIG. 1A.

[0009] FIG. 1E is close-up view of a portion of the top view the container loading apparatus shown in FIG. 1D.

[0010] FIG. 1F is a rear view of the container loading apparatus shown in FIG. 1A.

[0011] FIG. 1G is a sectional view of the container loading apparatus shown in FIG. 1A taken along the lines A-A shown in FIG. 1F.

[0012] FIG. 1H is a right side view of the container loading apparatus shown in FIG. 1A.

[0013] FIG. 2A is a sectional view of the container loading apparatus shown in FIG. 1A taken along the lines A-A shown in FIG. 1F but with the apparatus transitioning to an unloading position.

[0014] FIG. 2B is a right side view of the container loading apparatus in the position shown in FIG. 2A.

[0015] FIG. 3A is a sectional view of the container loading apparatus shown in FIG. 1A taken along the lines A-A shown in FIG. 1F but with the apparatus transitioning to an unloading position.

[0016] FIG. 3B is a right side view of the container loading apparatus in the position shown in FIG. 3A.

[0017] FIG. 4A is a sectional view of the container loading apparatus shown in FIG. 1A taken along the lines A-A shown in FIG. 1F but with the apparatus in an unloading position.
[0018] FIG. 4B is a right side view of the container loading apparatus in the unloading position shown in FIG. 4A.

[0019] FIG. 5 is a right side view of the container loading apparatus in the unloading position showing a maximum envelope space of movement for the example embodiment shown in FIG. 1A.

[0020] FIG. 6 is a right rear isometric view of one example embodiment of a container unloading apparatus consistent with the present disclosure wherein the apparatus is in a loading position.

[0021] FIG. 7 is a right side view of the container loading apparatus shown in FIG. 6.

[0022] FIG. 8 is a right side view of the container loading apparatus shown in FIG. 6 with the apparatus transitioning to an unloading position.

[0023] FIG. 9 is a right side view of the container loading apparatus shown in FIG. 6 with the apparatus transitioning to an unloading position.

[0024] FIG. 10 is a right side view of the container loading apparatus shown in FIG. 6 with the apparatus in an unloading position.

DETAILED DESCRIPTION

[0025] The present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The examples described herein may be capable of other embodiments and of being practiced or being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting as such may be understood by one of skill in the art. Throughout the present description, like reference characters may indicate like structure throughout the several views, and such structure need not be separately discussed. Furthermore, any particular feature(s) of a particular exemplary embodiment may be equally applied to any other exemplary embodiment(s) of this specification as suitable. In other words, features between the various exemplary embodiments described herein are interchangeable, and not exclusive.

[0026] An apparatus for emptying the contents of a container consistent with embodiments of the present disclosure may include a base, a frame, and a vertical lift. The lift may include a drive portion fixed to the base and a linearly movable member pivotally coupled to the frame. The drive portion may be configured for driving the linearly movable member for movement relative to the drive portion and the base along a substantially linear axis. The lift may also include a pivot arm having a first end coupled to the base and a second end pivotally coupled to the frame. As the drive portion drives the linearly movable member along the substantially linear axis the frame moves upwardly and rotates relative to the base. The frame may thus be moved by the vertical lift from a loading position where a container may be loaded onto the frame, e.g., through an open end of the frame, to an unloading position where the container may be unloaded, e.g. from an open top of the frame. Due to the vertical and rotational movement of the frame by the vertical lift, as the frame moves from the loading position to the unloading position the frame remains in a compact envelope space. This allows use of apparatus consistent with the present disclosure at locations where space is limited and/or in a manner that efficiently uses existing space.

[0027] FIGS. 1A-1H illustrate one example embodiment of **100** consistent with the present disclosure. In general, the illustrated system **100** includes a base **102**, a frame **104**, and first **106a** and second **106b** vertical lifts. Each of the vertical lifts **106a**, **106b** includes a drive portion **108a**, **108b**, respectively, fixed to the base **102**, and a linearly movable member **110a**, **110b** pivotally coupled to the frame **104**. Each drive portion **108a**, **108b** is configured for driving an associated linearly movable member **110a**, **110b** for movement along a substantially linear axis A1 relative to the drive portion **108a**, **108b** and the base **102**. Each vertical lift **106a**, **106b** also includes a pivot arm **112a**, **112b**. As shown in FIGS. 1D and 1E, each pivot arm **112a**, **112b** is a rigid elongate member and has a first end **114** coupled to the base **102** and a second end **116** pivotally coupled to the frame **104**.

[0028] In FIGS. 1A-1H the frame **104** is shown in a loading position. As each drive portion **108a**, **108b** drives the associated linearly movable member **110a**, **110b** along the substantially linear axis A1, the frame **104** moves upwardly and rotates relative to the base **102** to an unloading position, as shown in FIGS. 4A-B. FIGS. 2A-2B, 3A-3B respectively show different positions of the frame **104** as the vertical lifts **106a**, **106b** transition the frame **104** from the loading position shown in FIGS. 1A-H to the unloading position shown in FIGS. 4A-4B.

[0029] The base **102** is a rigid structure, e.g., formed from metallic components, with a bottom **118** (FIG. 1C) configured to lay flat on a floor surface **120**. The base **102** may be fixedly attached to a portion of the floor surface **120** (i.e., the base **102** may be bolted the floor surface **120**). In some embodiments, the base **102** may be removably positioned on the floor surface **120** to allow for portability of the system **100**. In the illustrated example embodiment and as shown particularly in FIG. 1D, the base **102** is generally rectangular with a front **122**, an open rear **124**, and left **126** and right sides **128**. The base **102** thus defines a cavity **130** for receiving the frame **104** when the frame **104** is in the loading position.

[0030] With reference to FIG. 1A, in the illustrated example embodiment, the frame **104** includes a platform **132** for supporting a container **134**, a front **136**, an open rear **138**, left **140** and right **142** sides, and an open top **144**. The frame **104** thus defines a cavity **146** for receiving the container **134** through the open rear end **138** and the container **134** may be supported within the cavity **146** on the platform **134**. The front **136** and left **140** and right **142** sides may extend to a height greater than a height of the container **134**.

[0031] Although specific embodiments of the base **102** and frame **104** are shown in the example embodiments, it will be recognized that the base **102** and the frame **104** may be provided in a variety of configurations. For example, the size, shape and dimensions of the base **102** and the frame **104** may be selected depending on the configuration of the container **134** to be loaded or emptied using the apparatus, the space where the apparatus where the apparatus is to be used, etc. The illustrated example embodiment is thus provided by way of illustration, not of limitation.

[0032] The vertical lifts **106a**, **106b** may be provided in a variety of configurations. Also, in the illustrated example embodiment, a first vertical lift **106a** is provided on left side **140** of the frame **104** and a second vertical lift **106b** is provided on the right side **142** of the frame **104**. It is to be understood, however, that an apparatus consistent with the present disclosure need not contain two vertical lifts **106a**, **106b** but may include any number of vertical lifts, including only one vertical lift. For simplicity and ease of explanation, the description herein may refer to the configuration and operation of only one of the illustrated vertical lifts, e.g. lift **106a** or **106b**. It is to be understood that the other of the illustrated vertical lifts **106a**, **106b** is configured in the same manner.

[0033] In the illustrated example embodiment, each of the vertical lifts **106a**, **106b** is configured as a linear actuator and the drive portion **108a**, **108b** includes an electric motor and the linearly movable member **110a**, **110b** is configured as an elongate generally rectangular lift arm **148a**, **148b**. The motor is configured, in a known manner, to drive the lift arm **148a**, **148b** along the linear axis A1 and relative to the base **102** and the drive portion **108a**, **108b**. Each drive portion **108a**, **108b** is disposed within, and is coupled to, a lift tower **150a**, **150b** that is fixed to the base **102**. The lift arm **148a**, **148b** is partially disposed within the lift tower **150a**, **150b** and extends from a top of the lift tower **150a**, **150b** when driven by the drive portion **108a**, **108b**. The lift tower **150a**, **150b** may be configured to block access to moving portions of the vertical lift **106a**, **106b** to prevent injury to a user.

[0034] In other embodiments, one or more of the vertical lifts **106a**, **106b** may include a hydraulic or pneumatic piston, and the drive portion **108a**, **108b** may be a piston of the hydraulic or pneumatic position and the linearly movable member **110a**, **110b** may be a piston rod of the hydraulic or pneumatic position. Also, the vertical lifts **106a**, **106b** need not include a lift tower **150a**, **150b**. Instead, the drive portion **108a**, **108b** of each of the vertical lifts **106a**, **106b** may be

coupled directly to the base **102**.

[0035] With particular reference to FIG. **1E**, in the illustrated example embodiment the lift arm **148b** is coupled to the frame **104** by an extension **152** extending between the lift arm **148b** and the frame **104**. A first end **154** of the extension **152** is fixed to a plate **156**, which is fixed to the lift arm **148b**. A second end **158** of the extension **152** is pivotally coupled to the frame **104** to allow the frame **104** to rotate relative to the lift arm **148b** and the base **102**. As each drive portion **108a**, **108b** drives the lift arm **148b** to move the frame **104** from the loading position to the unloading position, the lift arm **148a**, **148b** lifts the frame **104** along the linear axis A1 while the pivotal connection between the lift arm **148a**, **148b** and the frame **104** allows the frame **104** to rotate relative to the lift arm **148a**, **148b** and the base **102**. A variety of configurations for pivotally coupling the linearly movable members **110a**, **110b** to the frame **104** are possible. For example, in some embodiments the second end **158** of the extension **152** may be fixed to the frame **104** and the first end **154** of the extension **152** may be pivotally coupled to the lift arm **148b**, e.g. to the plate **156**.

[0036] The pivot arms **112a**, **112b** may be coupled between the base **102** and the frame **104** in a variety of configurations. With continued reference to FIG. **1E**, in the illustrated embodiment the pivot arm **112b** is coupled to the base **102** through an extension **160** coupled to a bracket **162** fixed to the lift tower **150b**. In some embodiments, however, the pivot arm **112b** may be coupled directly to the lift tower **150b** or the base **102** without use of an extension or a bracket **162**.

[0037] In the illustrated example embodiment, the extension **160** is fixed to the base **102** through the bracket **162**, and the first end **114** of the pivot arm **112b** is pivotally coupled to an end **164** of the extension **160**. A second end **116** of the pivot arm **112b** is pivotally coupled to the frame **104**. As each drive portion **108a**, **108b** drives the lift arm **148a**, **148b** to move the frame **104** from the loading position to the unloading position, the lift arm **148a**, **148b** lifts the frame **104** along the linear axis A1 while the pivotal connection between the pivot arms **112a**, **112b** and the frame **104** constrains the movement of the frame **104** to cause rotation of the first ends **114** of the pivot arms **112a**, **112b** relative to the extension and the base **102** and rotation of frame **104** relative to the second ends **116** of the pivot arms **112a**, **112b** and the base **102**.

[0038] FIG. **1G** is a side sectional view taken along the lines A-A in FIG. **1F**, and FIG. **1H** is a side view of the apparatus **100**. In the illustrated example embodiment, the linearly movable member **110a**, **110b** is pivotally coupled to the frame **104**, e.g. by the extension **152**, for rotation about a first pivot point P1, the second end **116** of the lift arm **148a** is pivotally coupled to the frame **104** for rotation about a second pivot point P2, and the first end **114** of the lift arm **148a** is coupled to the base **102**, e.g. by the extension **160** and bracket **162**, for rotation about a third pivot point P3.

[0039] The positions of the pivot points P1, P2, P3 on the frame **104** may be varied depending on the desired motion of the frame **104** when moving between the loading and unloading positions. In the illustrated example embodiment, the first pivot point P1 is positioned at a linear distance D1 from the front **136** of the frame **104** and at height H1 from the bottom of the frame **104**. The second pivot point is positioned at a linear distance D2 from the front **136** of the frame **104** and at a height H2 from the bottom of the frame **104**. The third pivot point is positioned at a linear distance D3 from the front **136** of the frame **104** and at a height H3 from the bottom of the frame **104**. In the illustrated example embodiment, the distance D1 is greater than the distance D2, which is greater than D3, i.e. $D1 > D2 > D3$. Also, height H3 is greater than the H2, which is greater than the height H1, i.e. $H3 > H2 > H1$. With this configuration the vertical lifts **106a**, **106b** may lift the frame **104** vertically while the pivot arms **112a**, **112b** cause forward rotation of the frame **104** to empty contents of the container **134** from the open top **144** of the frame **104**. In some embodiments, the relative positions of the first P1, second P2 and/or third P3 pivot points may be modified to cause a different motion of the frame **104** relative to the base **102**.

[0040] FIG. **2A** is a side sectional view taken along the lines A-A in FIG. **1F**, but when the frame **104** is an intermediate position during transition from the loading position shown in FIG. **1F** to the unloading position shown in FIGS. **4A** and **4B**. FIG. **2B** is a side view of the apparatus **100** in the

intermediate position shown in FIG. 2A. In the illustrated position the vertical lifts **106a**, **106b** have moved the frame **104** vertically along the linear axis A1 from the position shown in FIG. 1G and the frame **104** has rotated forwardly, i.e. toward the front **122** of the base **102**, about the first pivot point P1. Also, the frame **104** has rotated forwardly about the second pivot point P2 and the pivot arm **112a** has rotated forwardly about the third pivot point P3. The frame **104** has thus been lifted and forwardly rotated by about 45 degrees.

[0041] FIG. 3A is a side sectional view taken along the lines A-A in FIG. 1F, but when the frame **104** is in a further intermediate position during transition from the loading position shown in FIG. 1F to the unloading position shown in FIGS. 4A and 4B. FIG. 3B is a side view of the apparatus **100** in the intermediate position shown in FIG. 3A. In the illustrated position, and compared to FIG. 2A, the vertical lifts **106a**, **106b**, have moved the frame **104** vertically along the linear axis A1 from the position shown in FIG. 2A and the frame **104** has further rotated about the first pivot point P1. Also, the frame **104** has rotated forwardly about the second pivot point P2 and the pivot arm **112a** has rotated forwardly about the third pivot point P3. The frame **104** has thus been lifted and forwardly rotated by about 90 degrees.

[0042] FIG. 4A is a side sectional view taken along the lines A-A in FIG. 1F, but when the frame **104** is in the unloading position. FIG. 4B is a side view of the apparatus in the unloading position shown in FIG. 4A. In the illustrated position, and compared to FIG. 3A, the vertical lifts **106a**, **106b** have moved the frame **104** vertically along the linear axis A1 and the frame **104** has further rotated about the first pivot point P1. Also, the frame **104** has rotated forwardly about the second pivot point P2 and the pivot arm **112a** has rotated rearwardly about the third pivot point P3. The frame **104** has thus been lifted and forwardly rotated by about 135 degrees. In this unloading position, the open top **144** of the frame **104** is downwardly oriented to allow the contents of a container **134** to empty from the open top **144** of the frame **104** and, for example, onto a receptacle **166** such as a platform, a conveyor or receiving container.

[0043] Transition of the frame **104** from the unloading position shown in FIGS. 4A-4B back to the loading position shown in FIGS. 1A-1H may proceed in a reverse order from that described above. A controller **168** (FIG. 1) may be coupled to the system **100** through a wired or wireless connection to provide control signals for energizing the drive portions **108a**, **108b** associated with each of the vertical lifts **106a**, **106b** to cause the vertical lifts **106a**, **106b** to move the frame **104** between the loading and unloading positions. The controller **168** may provide the control signals to the system **100** autonomously, or in response, at least in part, to operator input.

[0044] Advantageously, due to the vertical and rotational movement of the frame **104** as the frame **104** moves from the loading position to the unloading position the frame **104** remains in a compact envelope space. FIG. 5 illustrates one example of the envelope space associated with the apparatus shown in FIGS. 1A-4B. It is to be understood, however, that the specific envelope space in any embodiment consistent with the present invention depends on the size of the apparatus, e.g. the size of the base **102** and the frame **104**. Also, the envelope space may be selectively modified in embodiments consistent with the present disclosure by modifying the position of first pivot point P1, second pivot point P2 and/or the third pivot point P3.

[0045] In the embodiment shown in FIG. 5, the base **102** has a width WB from the rear of the base **102** to the front of the base and a height HB from the bottom of the base **102** within which the frame **104** is contained when the frame **104** is in the loading position as shown in FIGS. 1A-1H. When the frame **104** is moved to the unloading position, the outermost point P0 of the frame **104** is at a width WF from the rear of the base **102** and the uppermost point PU of the frame **104** is positioned at a height HF from the bottom of the frame **104**. In some embodiments, the height HF may be less than about 190% of the height HB of the base **102** and the width WF may be less than about 150% of the width WB of the base **102**. In one embodiment consistent with the present disclosure, the width WB may be about 77.5 inches, the width WF may be about 104.625 inches, the height HB may be about 66.5625 inches, the height HF may be about 114.625. In some

embodiments, therefore, the height HF may thus be about 172% of the height HB of the base **102** and the width WF may be about 135% of a width WB of the base **102**.

[0046] A container unloading apparatus consistent with the present disclosure may be provided in a variety of configurations. FIGS. **6-10**, for example, illustrate another example embodiment **100a** of a container unloading apparatus consistent with the present disclosure.

[0047] In general, the illustrated apparatus **100a** includes a base **102a**, a frame **104a**, and first **106a** and second **106b** vertical lifts. Each of the vertical lifts **106a**, **106b** includes a drive portion fixed to the base **102a**, and a linearly movable member **110a**, **110b** pivotally coupled to the frame **104a**. Each drive portion is configured for driving an associated linearly movable member **110a**, **110b** for movement along a substantially linear axis relative to the drive portion and the base **102a**. Each vertical lift **106a**, **106b** also includes a pivot arm **112a**, **112b**. Each pivot arm **112a**, **112b** is a rigid elongate member and has a first end **114** coupled to the base **102a**, e.g. through the bracket **162**, and a second end **116** pivotally coupled to the frame **104a**.

[0048] In FIGS. **6** and **7** the frame **104a** is shown in a loading position. As each drive portion drives the associated linearly movable member **110a**, **110b** along the substantially linear axis, the frame **104a** moves upwardly and rotates relative to the base **102a** to an unloading position, as shown in FIG. **10**. FIGS. **8** and **9** respectively show different positions of the frame **104a** as the vertical lifts **106a**, **106b** transition the frame **104a** from the loading position shown in FIGS. **6** and **7** to the unloading position shown in FIGS. **10**.

[0049] The base **102a**, the frame **104a**, the first **106a** and second **106b** vertical lifts, the linear moveable members **110a**, **110b**, the drive portions, and the pivot arms **112a**, **112b** of the apparatus **100a** are coupled and operate to move the frame **104a** about pivot points P1, P2 and P3, as described above in connection with the apparatus **100**, which will not be repeated for ease of explanation. In the illustrated example embodiment, the unloading position is reached when the frame **104a** has been rotated to substantially 90 degrees from the loading position, as shown in FIG. **10**, and also in FIGS. **3A** and **3B** with respect to the apparatus **100**.

[0050] The apparatus **100a** also includes an extension frame **602**, a conveyor **604**, and a ramp **612**. The extension frame **602** has rigid sides defining a central opening **606** between the rigid sides. In general, the sides of the extension frame **602** are sized to receive the container **134** with at least a portion of the perimeter of the container **134** positioned adjacent shelf portions of the sides of the extension frame **602** at the front of the extension frame **602**. The central opening **606** of the extension frame **602** may thus be positioned adjacent the open end **608** of the container **134**. In some embodiments, the extension frame **602** may have fixed dimensions corresponding to the dimensions of the container **134**. In some embodiments, the extension frame **602** may have adjustable dimensions to allow adjustment of the extension frame **602** for use with containers **134** of different dimensions. The rear of the extension frame **602** is positioned adjacent the conveyor **604**, such that the extension frame **602** is positioned between a top surface **610** of the conveyor **604** the open end **608** of the container **134**.

[0051] The extension frame **602** may be slidably movable with respect to the conveyor **604** and the remainder of the frame **104a** and may be coupled to an extension frame drive portion **702** for driving the extension frame **602** and the container **134** between a non-extended position relative to the remainder of the frame **104a**, shown in FIGS. **6-9**, and an extended position shown in FIG. **10** relative to the remainder of the frame **104a**. In the non-extended position shown in FIGS. **6-9**, the central opening **606** of the extension frame **602** and the open end **608** of the container **134** are aligned adjacent to the conveyor **604**. In the extended position shown in FIG. **10**, the extension frame **602** and the container **134** are extended forwardly with respect to the frame **104a**, the conveyor **604** and the base **102a** to allow emptying of the contents of the container **134**. The conveyor **604** may be coupled to a conveyor drive portion **704** for driving the conveyor **604** for rotational movement about a conveyor frame. The extension frame drive portion **702** and the conveyor drive portion **704** may be known, hydraulic, electric and/or pneumatic drive systems, e.g.,

pistons and/or motors.

[0052] The ramp **612** has a bottom end **708** pivotally coupled to the frame **104a** and extends upwardly on the frame **104a** relative to the conveyor **602** when the apparatus **100a** is in a loading position. A top end **710** of the ramp **612** is not connected to the frame **104a**. A ramp drive portion **706** causes the ramp **612** to pivot relative to the frame **104a** at the bottom end **708** of the ramp **612**, causing the top end **710** of the ramp **612** to pivot away from the frame **104a** (as shown for example in FIGS. **9** and **10**). The ramp drive portion **706** may be a known, hydraulic, electric and/or pneumatic drive system, e.g., pistons and/or motors.

[0053] With reference to FIGS. **8** and **9**, as the apparatus **100a** moves the frame **104a** to the unloading position shown in FIG. **10**, the open end **608** of the container **134** is supported by the extension frame **602** with the contents of the container **134** passing through the open end **608** of the container **134** and the central opening **606** in the extension frame **602** to rest on the top surface **610** of the conveyor **604**. The extension frame drive portion **702** then drives the extension frame **602** and the container **134** supported thereon toward the extended position, the conveyor drive portion **704** drives the conveyor **604** to rotate, and the ramp drive portion **706** rotates the ramp **612** around the bottom end **708** of the ramp **612**.

[0054] The container **134** remains supported on the extension frame **602** with the open end **608** of the container **134** aligned with the central opening **606** of the extension frame **602** when the extension frame **602** and the container **134** are in the extended position. The contents of the container **134** thus fall from the container **134** and are guided onto a receptacle **166** by the ramp **612** and/or are conveyed by the conveyor **604** onto the ramp **612**, which guides them onto the receptacle **166**. In some embodiments, the rotational speed of the conveyor **604** may be substantially the same as the speed at which the extension frame **602** is driven from the non-extended position to the extended position to resist shearing of the contents of the container **134**. Advantageously, the frame **104a** remains in a compact envelope space as the frame **104a** moves from the loading position to the unloading position, as described above in connection with FIG. **5**.

[0055] Consistent with one aspect of the present disclosure there is provided an apparatus for emptying the contents of a container including: a base; a frame; and a lift. The lift includes a drive portion fixed to the base, a linearly movable member pivotally coupled to the frame, the drive portion configured for driving the linearly movable member for movement relative to the base along a substantially linear axis, and a pivot arm having a first end coupled to the base and a second end pivotally coupled to the frame. As the drive portion drives the linearly movable member along the substantially linear axis the frame moves upwardly and rotates relative to the base to transition from a loading position to an unloading position.

[0056] Consistent with another aspect of the present disclosure there is provided an apparatus for emptying the contents of a container including: a base; a frame; and a lift. The lift includes a drive portion fixed to the base, a linearly movable member pivotally coupled to the frame at a first pivot point, the drive portion configured for driving the linearly movable member for movement relative to the base along a substantially linear axis, and a pivot arm having a first end coupled to the base at a third pivot point and a second end pivotally coupled to the frame at a second pivot point. As the drive portion drives the linearly movable member along the substantially linear axis the frame moves upwardly and rotates relative to the base to transition from a loading position to an unloading position. When the frame is in the loading position the first pivot point is positioned at a first linear distance, $D1$, from a front of the frame and at a first height, $H1$, from a bottom of the frame, the second pivot point is positioned a second linear distance, $D2$, from the front of the frame and at a second height, $H2$ from the bottom of the frame, and the third pivot point is positioned at a third linear distance, $D3$, from the front of the frame and a third height, $H3$ from the bottom of the frame, wherein $D1 > D2 > D3$ and $H1 < H2 < H3$. As the frame moves from the loading position to the unloading position, the frame remains in an envelope space having a height less than 190% of a height of the base and a width less than 150% of a width the base.

[0057] In some embodiments the frame of the apparatus may include an extension frame slidably movable between a non-extended position and an extended position. In some embodiments, the frame may further include a conveyor positioned adjacent the extension frame.

[0058] Consistent with another aspect of the disclosure, there is provided an apparatus for emptying the contents of a container a base; a frame comprising an extension frame slidably movable between a non-extended position and an extended position; and a lift coupled to the frame for moving the frame between a loading position and an unloading position. In some embodiments, the apparatus may further include a conveyor positioned adjacent the extension frame. The conveyor is optional and may not be included in some embodiments.

[0059] Unless otherwise stated, use of the word “substantially” may be construed to include a precise relationship, condition, arrangement, orientation, and/or other characteristic, and deviations thereof as understood by one of ordinary skill in the art, to the extent that such deviations do not materially affect the disclosed methods and systems. The term “coupled” as used herein refers to any connection, coupling, link, or the like by which signals carried by one system element are imparted to the “coupled” element. Such “coupled” devices, or signals and devices, are not necessarily directly connected to one another and may be separated by intermediate components or devices that may manipulate or modify such signals. Likewise, the terms “connected” or “coupled” as used herein in regard to mechanical or physical connections or couplings is a relative term and does not require a direct physical connection. Elements, components, modules, and/or parts thereof that are described and/or otherwise portrayed through the figures to communicate with, be associated with, and/or be based on, something else, may be understood to so communicate, be associated with, and or be based on in a direct and/or indirect manner, unless otherwise stipulated herein.

[0060] Throughout the entirety of the present disclosure, use of the articles “a” and/or “an” and/or “the” to modify a noun may be understood to be used for convenience and to include one, or more than one, of the modified noun, unless otherwise specifically stated. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The phrase “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0061] Spatially relative terms, such as “beneath,” “below,” “upper,” “lower,” “above”, “left”, “right” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the drawings. These spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation shown in the drawings. For example, if the device in the drawings is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0062] Although the terms “first,” “second,” “third” etc. may be used to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections are not to be limited by these terms as they are used only to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer, or section could be termed a second element, component, region, layer, or section without departing from the scope and teachings of the present invention.

[0063] While the principles of the invention have been described herein, it is to be understood by those skilled in the art that this description is made only by way of example and not as a limitation as to the scope of the invention. Other embodiments are contemplated within the scope of the present invention in addition to the exemplary embodiments shown and described herein.

Modifications and substitutions by one of ordinary skill in the art are considered to be within the scope of the present invention, which is not to be limited except by the following claims.

Claims

1. An apparatus for emptying the contents of a container comprising: a base; a frame; and a lift comprising: a drive portion fixed to the base, a linearly movable member pivotally coupled to the frame, the drive portion configured for driving the linearly movable member for movement relative to the base along a substantially linear axis, and a pivot arm having a first end coupled to the base and a second end pivotally coupled to the frame, whereby as the drive portion drives the linearly movable member along the substantially linear axis the frame moves upwardly and rotates relative to the base to transition from a loading position to an unloading position.
2. The apparatus of claim 1, wherein the frame comprises a platform for supporting a container and an open end for receiving a container on the support.
3. The apparatus of claim 1, wherein the lift comprises a linear actuator and the drive portion is an electric motor and the linearly movable member is a lift arm.
4. The apparatus of claim 1, wherein the lift comprises a hydraulic cylinder and the drive portion is a hydraulic piston and the linearly movable member is a piston rod of the hydraulic cylinder.
5. The apparatus of claim 1, wherein the lift comprises a pneumatic cylinder and the drive portion is a pneumatic piston and the linearly movable member is a piston rod of the pneumatic cylinder.
6. The apparatus of claim 1, wherein the linearly movable member is pivotally coupled to the frame by an extension extending between the linearly movable member and the frame, a first end of the extension being fixed to the linearly movable member and a second end of the extension being pivotally coupled to the frame.
7. The apparatus of claim 1, wherein the first end of the pivot arm is pivotally coupled to the base.
8. The apparatus of claim 1, wherein the pivot arm is coupled to the base by an extension, a first end of the extension being fixed to the base and a second end of the extension being pivotally coupled to the a first end of the pivot arm.
9. The apparatus of claim 1, wherein the drive portion is disposed in a lift column and the pivot arm is coupled to a bracket extending from a side of the lift column.
10. The apparatus of claim 1, wherein the pivot arm is coupled to the bracket by an extension, the extension having a first end fixed to the bracket and a second end pivotally coupled to the first end of the pivot arm.
11. The apparatus of claim 1, wherein the linearly movable member is pivotally coupled to the frame at a first pivot point, the second end of the pivot arm is pivotally coupled to the frame at a second pivot point, and the first end of the pivot arm is pivotally coupled to the base at a third pivot point.
12. The apparatus of claim 11, wherein when the frame is in the loading position the first pivot point is positioned at a first linear distance, $D1$, from a front of the frame and at a first height, $H1$, from a bottom of the frame, and wherein the second pivot point is positioned at a second linear distance, $D2$, from the front of the frame and a second height, $H2$, from the bottom of the frame, and wherein $D1 > D2$ and $H1 < H2$.
13. The apparatus of claim 12, wherein the third pivot point is positioned at a third linear distance, $D3$, from a front of the frame and at a third height, $H3$, from a bottom of the frame, and wherein $D1 > D2 > D3$ and $H1 < H2 < H3$.
14. The apparatus of claim 1, wherein the frame has an open rear end for receiving a container in the loading position of the frame and an open top for emptying the container in the unloading position of the frame.
15. The apparatus of claim 1, wherein as the frame moves from the loading position to the unloading position, the frame remains in an envelope space having a height less than 190% of a

height of the base and a width less than 150% of a width the base.

16. The apparatus of claim 1, wherein the envelope space has a height of about 172% of a height of the base and a width of about 135% of a width of the base.

17. The apparatus according to claim 1, wherein the frame comprises an extension frame supporting the container, the container being slidably movable with the extension frame between a non-extended position and an extended position.

18. The apparatus according to claim 17, wherein the frame further comprises a conveyor positioned adjacent the extension frame.

19. The apparatus according to claim 17, the apparatus further comprising a ramp pivotally coupled to the frame adjacent the conveyor.

20. An apparatus for emptying the contents of a container comprising: a base; a frame; and a lift comprising: a drive portion fixed to the base, a linearly movable member pivotally coupled to the frame at a first pivot point, the drive portion configured for driving the linearly movable member for movement relative to the base along a substantially linear axis, and a pivot arm having a first end coupled to the base at a third pivot point and a second end pivotally coupled to the frame at a second pivot point, whereby as the drive portion drives the linearly movable member along the substantially linear axis the frame moves upwardly and rotates relative to the base to transition from a loading position to an unloading position, wherein when the frame is in the loading position the first pivot point is positioned at a first linear distance, $D1$, from a front of the frame and at a first height, $H1$, from a bottom of the frame, the second pivot point is positioned a second linear distance, $D2$, from the front of the frame and at a second height, $H2$ from the bottom of the frame, and the third pivot point is positioned at a third linear distance, $D3$, from the front of the frame and a third height, $H3$ from the bottom of the frame, and wherein $D1 > D2 > D3$ and $H1 < H2 < H3$, and wherein as the frame moves from the loading position to the unloading position, the frame remains in an envelope space having a height less than 190% of a height of the base and a width less than 150% of a width the base.
